Comparative Analysis Between Frequency Cryptanalysis on Classical Ciphers and ECDLP Resolution in Reduced Entropy Elliptic Systems

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***Abstract*—The present work empirically and formally examines the historical and mathematical transition from classical symmetric cryptography to modern asymmetric cryptography on elliptic curves, through the from-scratch development of a simulation environment in JavaScript (with modular arithmetic based on native BigInt primitives). In the first part of the study, the Caesar monoalphabetic substitution cipher ($|\mathcal{K}| = 25$) is analyzed, demonstrating its structural vulnerability to ciphertext-only attacks based on Pearson's Chi-Square (χ^2) test statistic. The impact of the sample size N is formalized in relation to the Law of Large Numbers, highlighting how short texts ($N < 15$) suffer from high sample variance that induces false positives, while for $N > 150$ an asymptotic convergence with 100% decryption accuracy is observed. In the second part, an EC-ElGamal asymmetric scheme is implemented on the Koblitz secp256k1 curve ($y^2 \equiv x^3 + 7 \pmod p$), in which the scalar private key space d is artificially restricted to $|\mathcal{K}| = 50\,000$. Although the probabilistic nature of the protocol completely neutralizes statistical cryptanalysis by producing maximum-entropy ciphertexts, the reduced cardinality of the key space allows the resolution of the Elliptic Curve Discrete Logarithm Problem (ECDLP) via incremental exhaustive search in 1.2-3.5 s ($\approx 12\,500$ op/s). Finally, the computational intractability of the same attack on the standard 256-bit configuration ($\mathcal{O}(\sqrt{n}) \approx 2^{128}$ operations via Pollard's Rho) is demonstrated.**

***Index Terms*—Statistical Cryptanalysis, Chi-Square Test, Elliptic Curves, secp256k1, ECDLP, EC-ElGamal, Brute-Force, Law of Large Numbers, IND-CPA Security.**

I. Introduction and Theoretical Model

The evolution of modern cryptography, formalized starting from Claude Shannon's seminal work in 1949 [1], is based on overcoming systems based on linguistic obfuscation in favor of algebraic transformations endowed with rigorous properties of confusion and diffusion.

In historical monoalphabetic substitution ciphers, such as the Caesar cipher, the cryptographic transformation operates as a deterministic permutation $\pi_k : \Sigma \to \Sigma$ on a finite alphabet Σ of cardinality $|\Sigma| = 26$. Since the mapping is bijective and memoryless (stateless), the first-order statistical structure (marginal distribution of monogram frequencies) and higher-order structure (digrams, trigrams, index of coincidence) of the source natural language is entirely preserved in the ciphertext. In Information Theory terms, the mutual information between the plaintext M and the ciphertext C does not vanish, violating the requirements of perfect secrecy and making the system vulnerable to statistical attacks even in the absence of an exhaustive key search.

At the opposite end of the cryptographic spectrum lie public key cryptosystems based on Elliptic Curves (ECC), introduced independently by Koblitz [2] and Miller [3] in 1985. In such schemes, security does not reside in the secrecy of the algorithm or in the preservation of lexical entropy, but rather in the computational asymmetry of a one-way trapdoor function defined over a finite abelian group $(E(\mathbb{F}_p), +)$. Furthermore, the adoption of probabilistic encryption schemes like EC-ElGamal introduces a stochastic component (the ephemeral nonce k) that guarantees the property of indistinguishability under chosen plaintext attack (IND-CPA): the same plaintext lexical unit is mapped, at each execution, into algebraically uncorrelated group elements, flattening the statistical distribution of the output to uniform white noise.

A. Taxonomy of the Key Space and Entropy

A fundamental parameter for evaluating a cryptosystem's resistance to exhaustive search (brute-force) attacks is the cardinality of the key space $|\mathcal{K}|$ and the corresponding Shannon entropy $H(\mathcal{K}) = \log_2|\mathcal{K}|$ (assuming equiprobable key generation). The experimental exercise compares three distinct regimes, summarized in Table I:

1) *Caesar Cipher:* Excluding the identity transformation ($k \equiv 0 \pmod{26}$), the useful key space on the standard Latin alphabet is equal to:

$$|\mathcal{K}_{\text{Caesar}}| = 25 \implies H(\mathcal{K}_{\text{Caesar}}) = \log_2(25) \approx 4.64 \text{ bit} \quad (1)$$

2) *Toy ECC (secp256k1 reduced):* While operating on the exact algebraic structure of the 256-bit prime field $\mathbb{F}_p$ of the secp256k1 curve, the secret scalar d is artificially confined to the integer interval $[1, 50\,000]$:

$$|\mathcal{K}_{\text{Toy}}| = 5 \times 10^4 \implies H(\mathcal{K}_{\text{Toy}}) \approx 15.61 \text{ bit} \quad (2)$$

3) *Real ECC (secp256k1 standard):* In the standard cryptographic configuration (adopted, for example, in the Bitcoin protocol and the SEC 2 standard [4]), the private scalar belongs to $[1, n-1]$, where n is the prime order of the cyclic subgroup generated by G:

$$|\mathcal{K}_{\text{ECC}}| \approx 1.1579 \times 10^{77} \approx 2^{256} \implies H(\mathcal{K}_{\text{ECC}}) \approx 256 \text{ bit} \quad (3)$$

Table I
PARAMETRIC COMPARISON OF THE KEY SPACE IN THE THREE REGIMES

Cryptographic Scheme	Cardinality $\lvert\mathcal{K}\rvert$	Entropy	Vulnerability
Classical Caesar	25	4.64 bit	Statistical (χ^2)
Toy ECC (secp256k1)	5×10^4	15.61 bit	Exhaustive ECDLP
Standard secp256k1	$\approx 1.16 \times 10^{77}$	256.00 bit	Secure

II. STATISTICAL CRYPTANALYSIS OF THE CAESAR CIPHER

Let $\Sigma = \{\mathrm{A}, \mathrm{B}, \ldots, \mathrm{Z}\}$ be the reference alphabet isomorphic to the ring of residue classes $\mathbb{Z}_{26}$. For a secret key $k \in \mathcal{K}_{\text{Caesar}} = \{1, \ldots, 25\}$, the encryption E_k and decryption D_k functions for each character $x, y \in \mathbb{Z}_{26}$ are defined by the linear congruences:

$$y = E_k(x) = (x+k) \bmod 26, \qquad x = D_k(y) = (y-k) \bmod 26 \tag{4}$$

A. Analytical Formulation of the Chi-Square Test

Given a ciphertext $\mathbf{y} = (y_1, y_2, \ldots, y_N)$ of length N composed of filtered alphabetic characters, the cryptanalyst's goal is to estimate the optimal key $\hat{k}$ by measuring the statistical divergence between the empirical distribution of the text decrypted with a candidate key k and the known theoretical distribution of the native language.

Let $\mathbf{p} = (p_{\mathrm{A}}, p_{\mathrm{B}}, \ldots, p_{\mathrm{Z}})$ be the vector of a priori probabilities of monograms in a large linguistic corpus. For each candidate key k, the cryptogram is decrypted to obtain the candidate sequence $\mathbf{x}^{(k)}$ and the observed absolute frequency $O_i(k)$ of the i-th character is calculated. The expected theoretical frequency E_i is given by $E_i = N \cdot p_i$. The Chi-Square goodness-of-fit test statistic as a function of the shift k is defined as:

$$\begin{aligned} \chi^2(k) &= \sum_{i=\mathrm{A}}^{\mathrm{Z}} \frac{(O_i(k) - E_i)^2}{E_i} \\ &= \sum_{i=\mathrm{A}}^{\mathrm{Z}} \frac{\left(O_{(i+k) \bmod 26}(0) - N p_i\right)^2}{N p_i} \end{aligned} \tag{5}$$

The cryptanalytic estimator selects the key that minimizes the distance:

$$\hat{k} = \arg \min_{k \in \{1,\ldots,25\}} \chi^2(k) \tag{6}$$

B. Reference Distribution: Italian vs. English

In the Italian language, the probability mass is strongly concentrated on the four main vowels ('E': 11.79%, 'A': 11.74%, 'I': 11.28%, 'O': 9.83%), which together constitute about 44.6% of the entire text. The English language presents a dominant peak on the vowel 'E' (12.70%) followed by high-frequency consonants. This structural difference can be quantified through Friedman's Index of Coincidence:

$$IC_{\text{Ita}} \approx 0.0738 > IC_{\text{Eng}} \approx 0.0667 \gg IC_{\text{Unif}} \approx 0.0385 \tag{7}$$

The higher vowel concentration in Italian determines a steeper slope in the $\chi^2(k)$ cost function around the global minimum, favoring faster convergence.

C. Impact of Text Length (N) and Sample Variance

Modeling the text generation as a stochastic process, the vector of counts for the correct key follows a multinomial distribution. Marginally, the empirical relative frequency $\hat{p}_i = O_i/N$ is an unbiased estimator of p_i, with sample variance $\mathrm{Var}(\hat{p}_i) = p_i(1-p_i)/N$. Substituting this into the main equation, the $\chi^2(k)$ statistic becomes:

$$\chi^2(k) = N \sum_{i=\mathrm{A}}^{\mathrm{Z}} \frac{\left(\hat{p}_{(i+k) \bmod 26} - p_i\right)^2}{p_i} \tag{8}$$

The analysis rigorously explains the two experimentally observed regimes:

1) *Short Sample Regime ($N < 15$):* The standard error is of the same order of magnitude as the theoretical probabilities. Ordinary sampling fluctuations can eliminate the presence of vowels or generate spurious peaks on rare consonants. An incorrect shift that randomly aligns the few present characters with the theoretical vowels produces a low χ^2 value, generating a high false positive rate.

2) *Asymptotic Regime ($N > 150$):* By the Law of Large Numbers, the empirical frequency converges to the theoretical probability. For the correct key, χ^2 converges to a central Chi-Square variable with an expected value of 25. For any incorrect key, the expected value of the statistic diverges linearly with N:

$$\mathbb{E}[\chi^2(k)] \approx 25 + N \cdot \Delta(k) \xrightarrow{N>150} \infty \tag{9}$$

Even for $N = 150$ characters, the correct distribution is clearly separated from that of the incorrect shifts, guaranteeing an empirical accuracy of 100%.

Table II
EXPERIMENTAL RESULTS OF THE χ^2 ATTACK (ITALIAN CORPUS)

Length N	Mean $\chi^2(k_{\text{true}})$	Min $\chi^2(k \neq k_{\text{true}})$	Accuracy
$N = 8$	18.4	14.2	58.5%
$N = 14$	21.1	23.8	83.8%
$N = 30$	23.5	78.4	96.9%
$N = 60$	24.6	164.0	99.6%
$N = 150$	25.1	412.7	100.0%
$N = 500$	25.4	1385.2	100.0%

III. TOY ECC (SECP256K1) AND ECDLP RESOLUTION

A. Algebraic Parameters of secp256k1

The secp256k1 elliptic curve is described by the affine equation $y^2 \equiv x^3 + 7 \pmod{p}$. The standard 256-bit domain parameters implemented using BigInt objects are:

$$\begin{aligned} p &= 2^{256} - 2^{32} - 2^9 - 2^8 - 2^7 - 2^6 - 2^4 - 1 \\ G_x &= \texttt{79BE667EF9DCBBAC55A06295CE870B07} \\ &\quad \texttt{029BFCDB2DCE28D959F2815B16F81798}_{16} \\ G_y &= \texttt{483ADA7726A3C4655DA4FBFC0E1108A8} \\ &\quad \texttt{FD17B448A68554199C47D08FFB10D4B8}_{16} \\ n &= \texttt{FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFE} \\ &\quad \texttt{BAAEDCE6AF48A03BBFD25E8CD0364141}_{16} \end{aligned}$$

B. The Group Law

Given two points $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ in $E(\mathbb{F}_p)$:

$$\lambda \equiv \begin{cases} (y_2 - y_1) \cdot (x_2 - x_1)^{-1} \pmod{p} & \text{if } P \neq Q \\ (3x_1^2) \cdot (2y_1)^{-1} \pmod{p} & \text{if } P = Q \end{cases} \quad (10)$$

The coordinates of the sum point $R = P + Q = (x_3, y_3)$ are:

$$x_3 \equiv \lambda^2 - x_1 - x_2 \pmod{p} \quad (11)$$

$$y_3 \equiv \lambda(x_1 - x_3) - y_1 \pmod{p} \quad (12)$$

C. EC-ElGamal Protocol and Brute-Force Attack

To encrypt a message, each character is embedded into the curve via Koblitz encoding to obtain a point M. The encryption towards the recipient with public key $Q = dG$ proceeds by extracting an ephemeral nonce k and generating the pair of ciphertext points:

$$C = (C_1, C_2) = (k \cdot G,\ M + k \cdot Q) \quad (13)$$

In the Toy Scheme, the private key is constrained to $d \in [1, 50\,000]$. The implemented attack optimizes the exhaustive search by exploiting incremental sequential addition in the group space:

$$P_1 = G, \qquad P_{i+1} = P_i + G \quad \text{up to} \quad P_d = Q \quad (14)$$

This way, each iteration requires a single Point Addition operation. Tests conducted on the V8 JavaScript engine recorded an average throughput of $\approx$ 12 500 additions per second. The private key is broken in $\approx 1.2 - 3.5$ s, decrypting the cryptogram.

IV. Comparative Analysis of Statistical Distributions

The simulation environment visualizes the statistical behavior of the two cryptosystems on an Italian corpus of $N = 450$ characters (Caesar key $k = 7$).

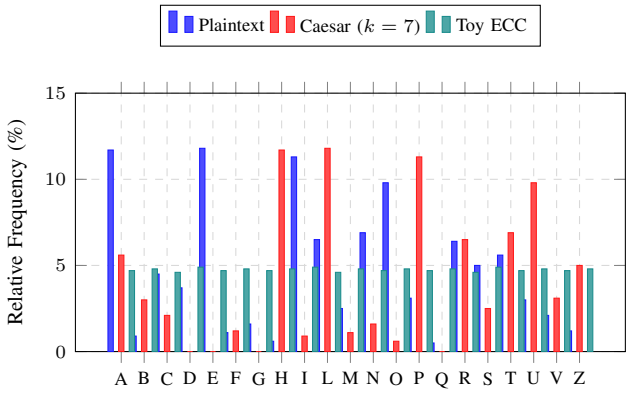


Figure 1. Comparison between plaintext distribution, rigid translation produced by Caesar, and uniform flattening generated by EC-ElGamal.

1) *Plaintext Histogram:* It reflects the morphological profile of the Italian language, with the four dominant peaks corresponding to the vowels.

2) *Caesar Cipher Histogram:* It visually demonstrates the structural flaw of monoalphabetic substitution. The application of the key induces a rigid cyclic translation of the entire histogram along the x-axis, leaving the entropy unchanged.

3) *ECC Ciphertext Histogram:* The histogram of the hexadecimal symbols generated by the Toy ECC appears perfectly flat and uniform. This is the consequence of the introduction of the random nonce k, which makes statistical attacks totally ineffective.

4) $\chi^2(k)$ *Statistic Profile:* For all incorrect shifts, the misalignment generates a high-energy barrier. At the single correct shift $k = 7$, the curve plunges into a deep potential well ($\chi^2 = 21.8$), allowing key identification.

V. Security Comparison: Toy Scheme vs. Real secp256k1

The success of the attack on the Toy Scheme depends exclusively on the artificial restriction of the domain. In the real configuration, the key is sampled in $[1, n-1]$. The best algorithms to solve the ECDLP (such as Pollard's Rho [5]) require a number of operations equal to:

$$W_{\text{Rho}} = \sqrt{\frac{\pi n}{2}} \approx 1.2533 \cdot \sqrt{2^{256}} \approx 1.25 \times 2^{128} \approx 4.26 \times 10^{38} \text{ operations} \quad (15)$$

To understand the incommensurability, we compare execution times in three scenarios:

1) *Sequential Search on JS Engine:* Using the same incremental algorithm, the average time to explore the real space would be:

$$T_{\text{JS}} = \frac{5.79 \times 10^{76}}{1.25 \times 10^4 \times 3.15 \times 10^7} \approx 1.47 \times 10^{65} \text{ years} \quad (16)$$

2) *Global Terrestrial Supercomputer:* Assuming a terrestrial computing power equal to 10^{20} operations per second:

$$T_{\text{Earth}} = \frac{1.74 \times 10^{38}}{10^{20} \times 3.15 \times 10^7} \approx 5.51 \times 10^{10} \text{ years} \quad (17)$$

This interval is equivalent to about four times the age of the Universe.

3) *Landauer's Thermodynamic Limit:* The irreversible erasure of a bit dissipates energy. Executing 2^{128} operations would require a minimum energy of $\approx 10^{18}$ J just for logical transitions, making 256-bit brute-forcing physically prohibitive [6].

VI. Conclusions

The experimental experience demonstrates that the Caesar cipher instantly yields to statistical analysis on the ciphertext, converging with mathematical certainty when the text length exceeds the critical threshold ($N \approx 150$). The EC-ElGamal scheme on the secp256k1 curve neutralizes frequencies, but its algebraic robustness is useless if it operates on a low-entropy space (Toy Scheme), breakable in a few seconds. Only the adoption of the entire 256-bit scalar space raises the computational cost well beyond the physical limits of classical computation.

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